CHARACTERIZATIONS OF RESOLVABILITY FOR A ONE DIMENSIONAL INTEGRAL NOETHERIAN SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file resolve.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction of the list of implications to the construction of the sequence of blowups, the factorization of a finite resolution through blowups of closed points with the associated ascending chain of pushforwards, and the termination through the ascending chain condition together with the universal property of blowing up.

Editorial note. The label lemma-resolve-curve names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source resolve.tex –label lemma-resolve-curve –start 4247 –end 4304 –local-id 28089 returned RESULT=OK (L28089 free, occupancy FREE, public interval clear). Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Finiteness of a resolution in dimension one and of the normalization for alterations. Varieties, the lemma that a modification of a one dimensional scheme is finite, and Same chapter, the lemma deducing finiteness of the normalization from a regular alteration. Supply the finiteness of the resolution used to build the ascending chain and the implication from a regular alteration to finiteness of the normalization.
- Ascending chain condition for coherent submodules of a coherent module. Cohomology of Schemes, the lemma that a coherent sheaf on a Noetherian scheme has the ascending chain condition on coherent submodules. Supplies the stabilization of the ascending chain of pushforwards of structure sheaves.
- Universal property of blowing up. Divisors, the lemma giving the universal property of the blowing up of an ideal, applied to the pullback of a maximal ideal which is invertible on the resolution. Supplies the factorization of the resolution through each blowup in the constructed sequence.

Source reference. Original lemma 01Y8: sources/stacks-chow/tags.txt, source lines 2547–2547. Prerequisite pointer invoked at proof line(s) 4290; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0806: sources/stacks-chow/tags.txt, source lines 10171–10171. Prerequisite pointer invoked at proof line(s) 4299; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BGG: sources/stacks-resolve/pinned-original-source.tex, source lines 3747–3747. Prerequisite pointer invoked at proof line(s) 4264; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AB7: sources/stacks-chow/tags.txt, source lines 13010–13010. Prerequisite pointer invoked at proof line(s) 4273; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let Y be a one dimensional integral Noetherian scheme. The following are equivalent*

(1) *there exists an alteration $X \to Y$ with X regular,*
(2) *there exists a resolution of singularities of Y,*
(3) *there exists a finite sequence $Y_n \to Y_{n-1} \to \ldots \to Y_1 \to Y$ of blowups in closed points with Y_n regular, and*
(4) *the normalization $Y^\nu \to Y$ is finite.*

Proof. The implications (3) $\Rightarrow$ (2) $\Rightarrow$ (1) are immediate. The implication (1) $\Rightarrow$ (4) follows from Lemma 0BGG. Observe that a normal one dimensional scheme is regular hence the implication (4) $\Rightarrow$ (2) is clear as well. Thus it remains to show that the equivalent conditions (1), (2), and (4) imply (3).

Let $f : X \to Y$ be a resolution of singularities. Since the dimension of Y is one we see that f is finite by Varieties, Lemma 0AB7. We will construct factorizations

$$X \to \ldots \to Y_2 \to Y_1 \to Y$$

where $Y_i \to Y_{i-1}$ is a blowing up of a closed point and not an isomorphism as long as Y_{i-1} is not regular. Each of these morphisms will be finite (by the same reason as above) and we will get a corresponding system

$$f_*\mathcal{O}_X \supset \ldots \supset f_{2,*}\mathcal{O}_{Y_2} \supset f_{1,*}\mathcal{O}_{Y_1} \supset \mathcal{O}_Y$$

where $f_i : Y_i \to Y$ is the structure morphism. Since Y is Noetherian, this increasing sequence of coherent submodules must stabilize (Cohomology of Schemes, Lemma 01Y8) which proves that for some n the scheme Y_n is regular as desired. To construct Y_i given Y_{i-1} we pick a singular closed point $y_{i-1} \in Y_{i-1}$ and we let $Y_i \to Y_{i-1}$ be the corresponding blowup. Since X is regular of dimension 1 (and hence the local rings at closed points are discrete valuation rings and in particular PIDs), the ideal sheaf $\mathfrak{m}_{y_{i-1}} \cdot \mathcal{O}_X$ is invertible. By the universal property of blowing up (Divisors, Lemma 0806) this gives us a factorization $X \to Y_i$. Finally, $Y_i \to Y_{i-1}$ is not an isomorphism as $\mathfrak{m}_{y_{i-1}}$ is not an invertible ideal. $\square$